



Network Setup and configuration using Packet Tracer

Lesson Description:

Students will be introduced to Packet Tracer to setup and configure a SOHO (small office home office) network. In addition, students will be introduced to the configuration of networking devices such as routers, switches, and end devices. Packet Tracer is a virtual networking software that allows students setup, configure and troubleshoot network domains

Length of Completion:

120 minutes

Level of Instruction:

6-8

Applicable First Principles &/or Concepts:

Please check the Principles or Concepts covered in this lesson.

Domain Principles

Domain Separation

Process Isolation

Resource Encapsulation

Modularity

Least Privilege

Abstraction

Data Hiding

Layering

Simplicity

Minimization

Cybersecurity Concepts

Defense in Depth

Availability

Confidentiality

Think Like an Adversary

Integrity

Keep it Simple

Resources that are Needed:

- 1) Windows PC
- 2) Packet Tracer Software (Download it from Cisco Netacad)

Accommodations Needed:

Major accommodation needed would be for visual impaired/disability students who would need additional assistance during the activity.

LEARNING OUTCOMES

LESSON LEARNING OUTCOMES:

- 1) Understand network fundamental concepts
- 2) Identify networking devices
- 3) Setup and configure a small network

LESSON DETAILS

Assessment:

The assessment for this activity is currently measure by the following:

1. Computer hardware and software activity
2. Activity demonstration
3. Instructor will observe all groups during activity completion

Extension Activities:

For this activity, there are not extensions

Differentiated Learning Opportunities:

Depending on the students' networking experience, some students will be given the opportunity to program the networking devices using CLI (command line interface) commands.

LESSON



Lesson 1 Details:

Warm Up:

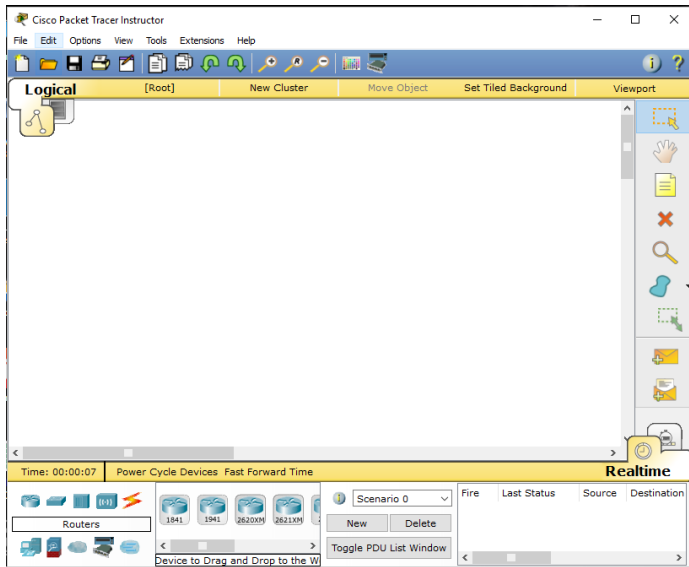
1. Presentation: Instructor will explain fundamental networking concepts, and the behavior of networking devices to achieve its functionality in a network environment. Students will be introduced to Packet Tracer (a networking simulation software) that allows students to setup simple and complex network topologies without using real hardware. Once the students complete the activity of setting up the SOHO network in the simulation software, they will be introduced to configure a 1841/1941 router and a 2960 switch to setup a small network environment. Students will verify their work by testing the connection from one device to another device within the same network domain.
2. Demonstration: Instructor will demonstrate student how to complete the networking activity.

Lesson:

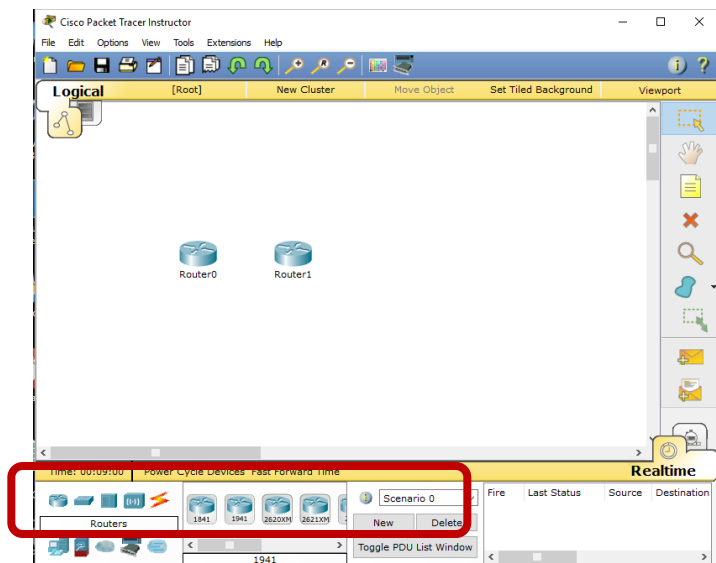
A SOHO (Small Office/Home Office) usually consists of a business that is privately owned or an individual who is self-employed, so the term usually refers to both a small office space as well as a small number of employees. Since the workload for these types of businesses are often primarily on the internet, they require a local area network (LAN), which means their network devices (hardware) is structured specifically for that purpose. A SOHO network can be a mixed network of wired and wireless computers just like other local networks. Since these types of networks are meant for businesses, they also tend to include printers and sometimes voice over IP (VoIP) and fax over IP technology.

In the following activity, we will setup and configure these devices using Packet Tracer.

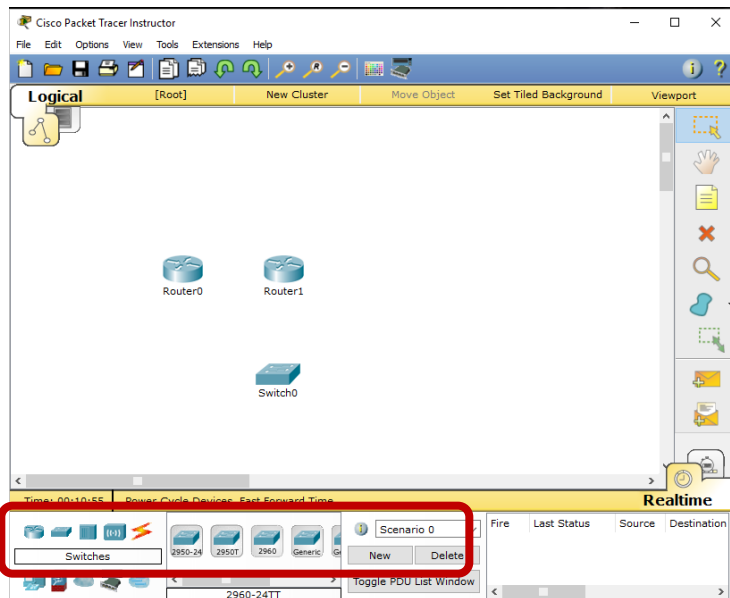
Step 1: Open Packet Tracer



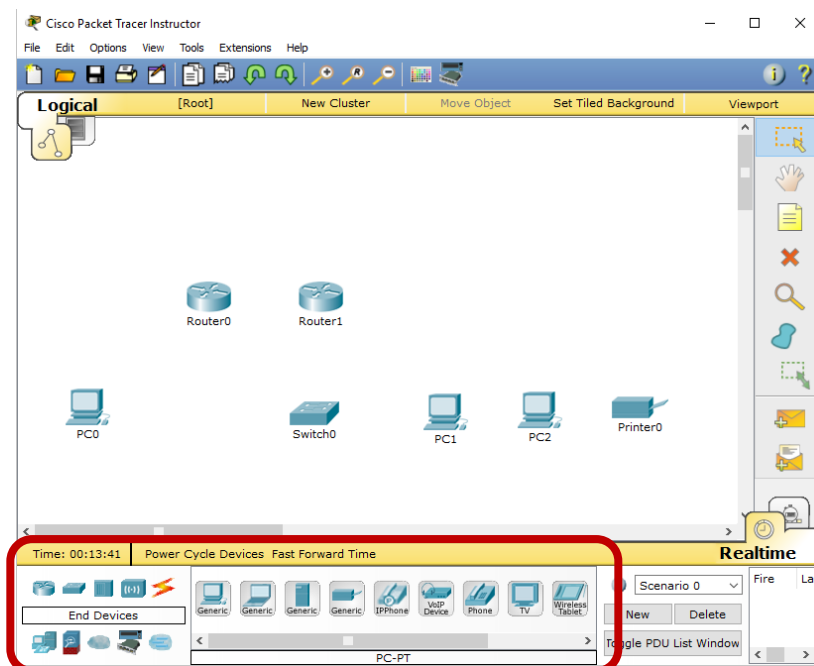
Step 2: Click on **Router** and add 2 routers (**1941**) to the workspace



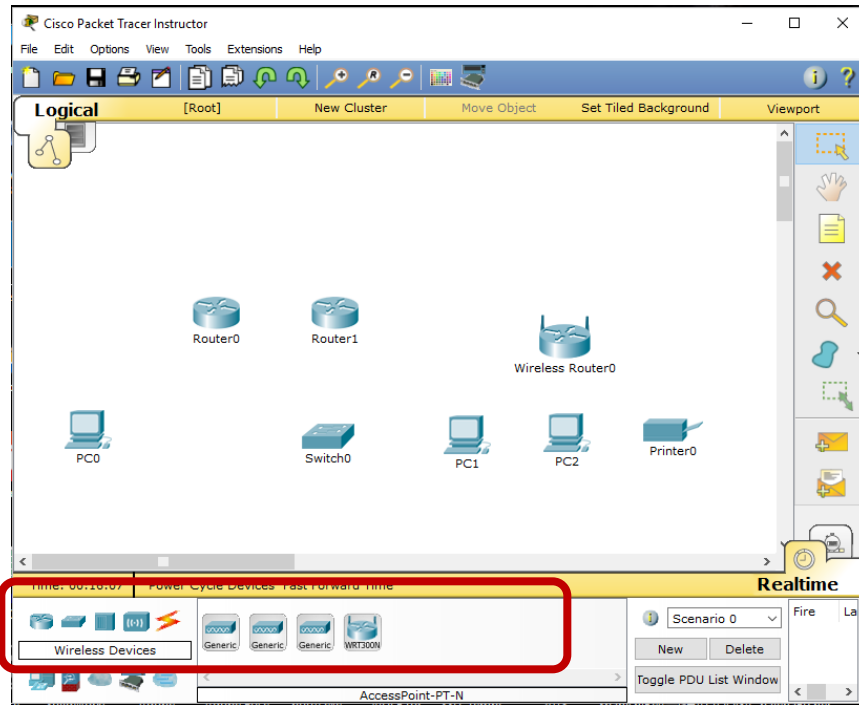
Step 3: Click on **Switches** and add 1 switch 2960 to the workspace



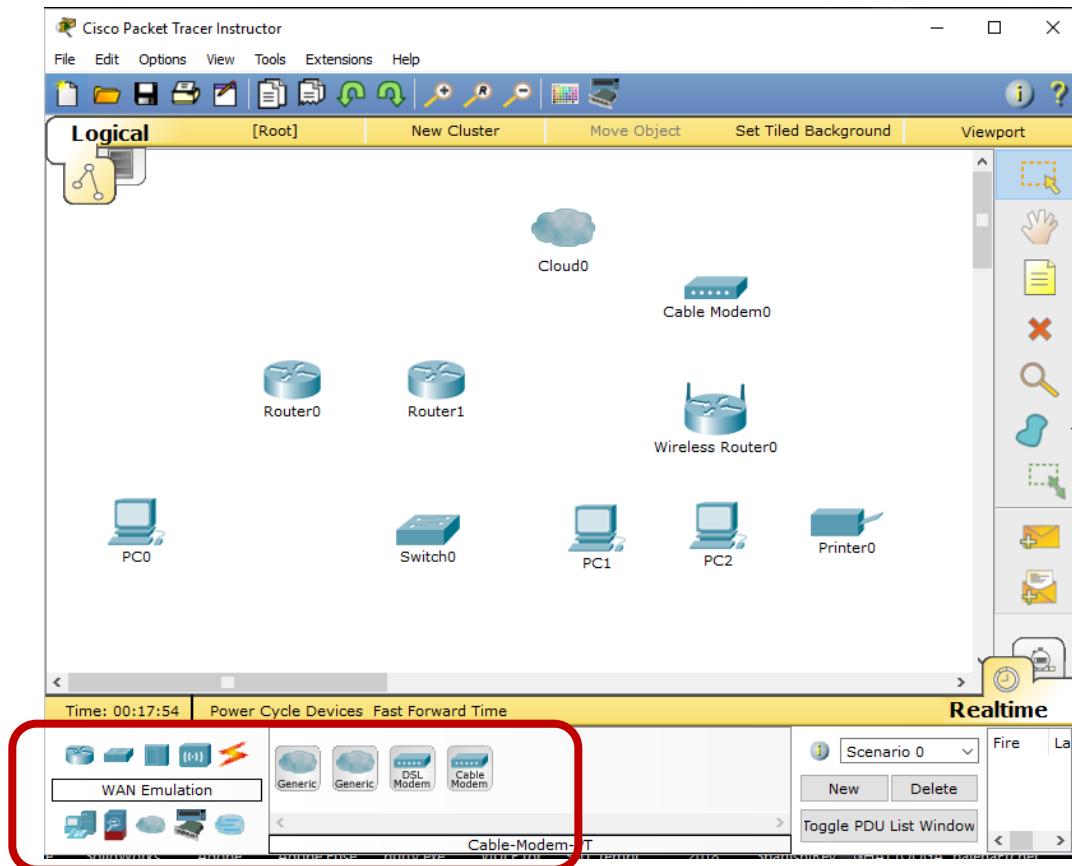
Step 4: Click on **End Devices** and add the 3 PCs and 1 printer



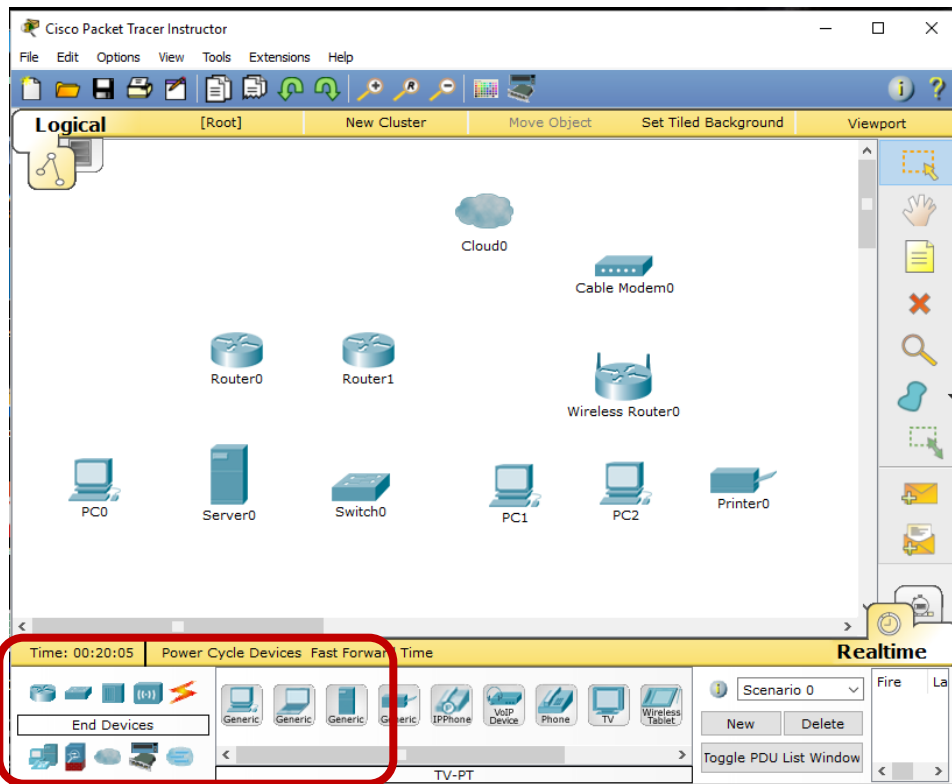
Step 5: Click on **Wireless Devices** and add a **Wireless Router WRT300N** to the workspace



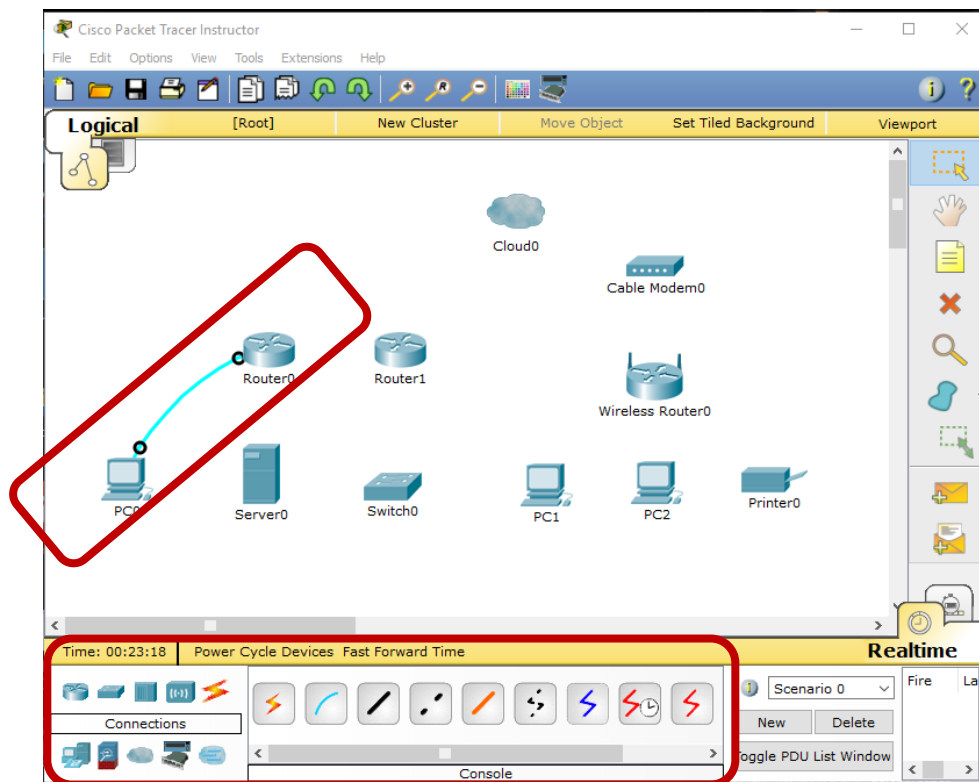
Step 6: Click on **WAN Emulation** and add **Generic Cloud** and a Cable Modem to the workspace



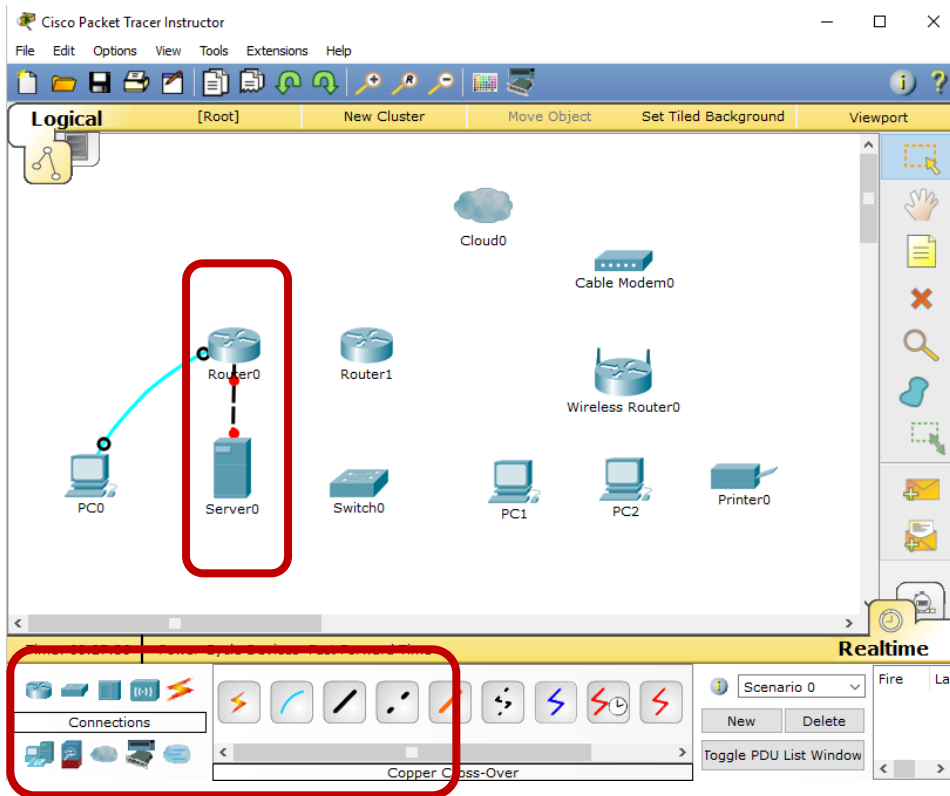
Step 7: Click on **End Devices** and add a **Server** to the workspace



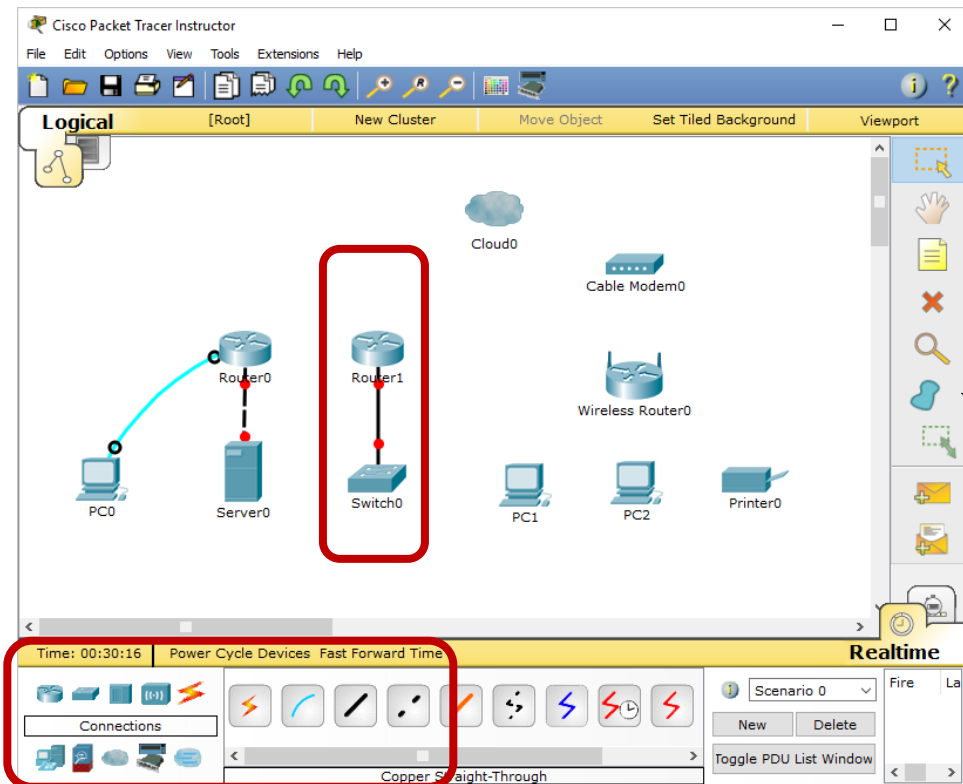
Step 8: Click on Connections, select **Console** to connect **PC0 (RS 232)** to **Router0 (Console)**



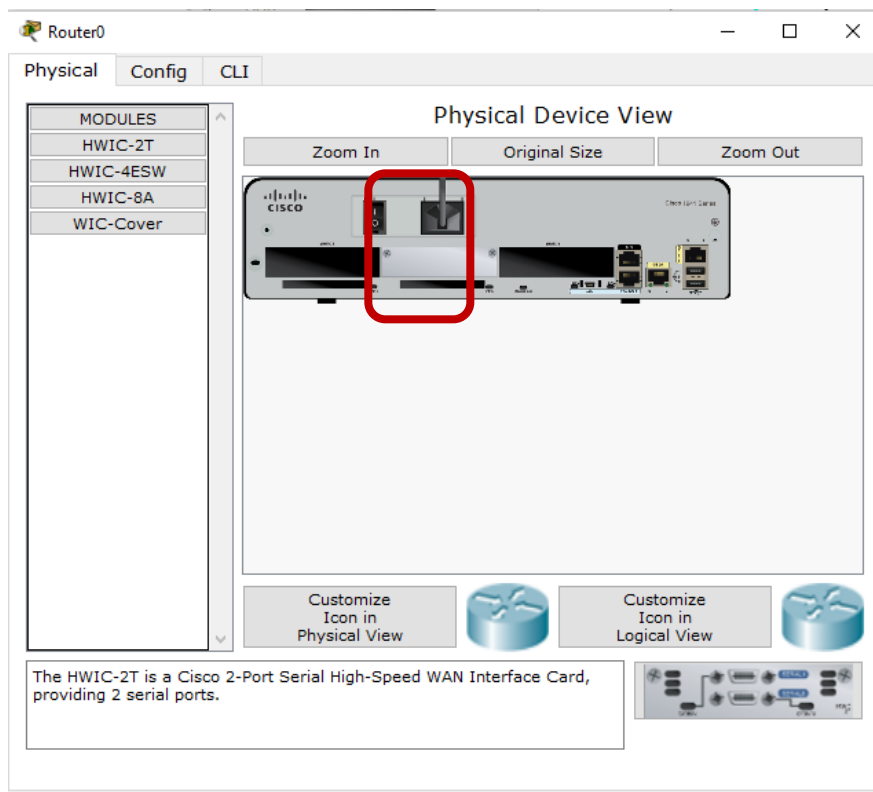
Step 9: Select **Copper Cross-over** to connect **Router0 (GigaEthernert0/0)** to **Server0 (FastEthernet0)**



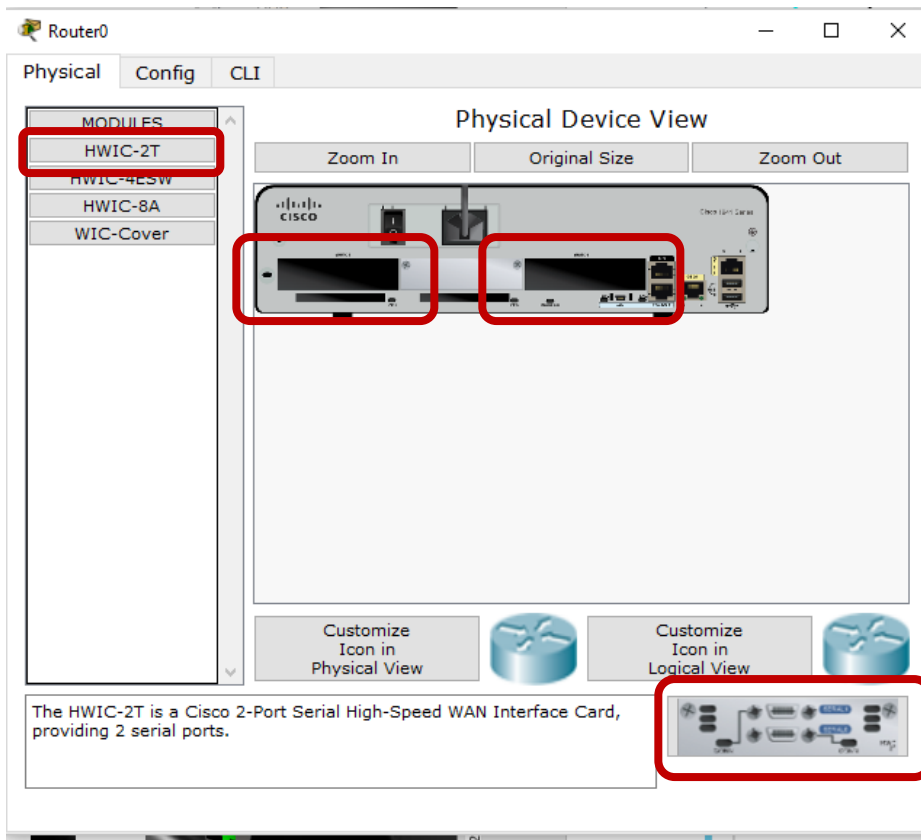
Step 10: Select **Copper Straight Through** to connect **Router1 (GigabitEthernet0/1)** to connect to **Switch0 (FastEthernet0/1)**



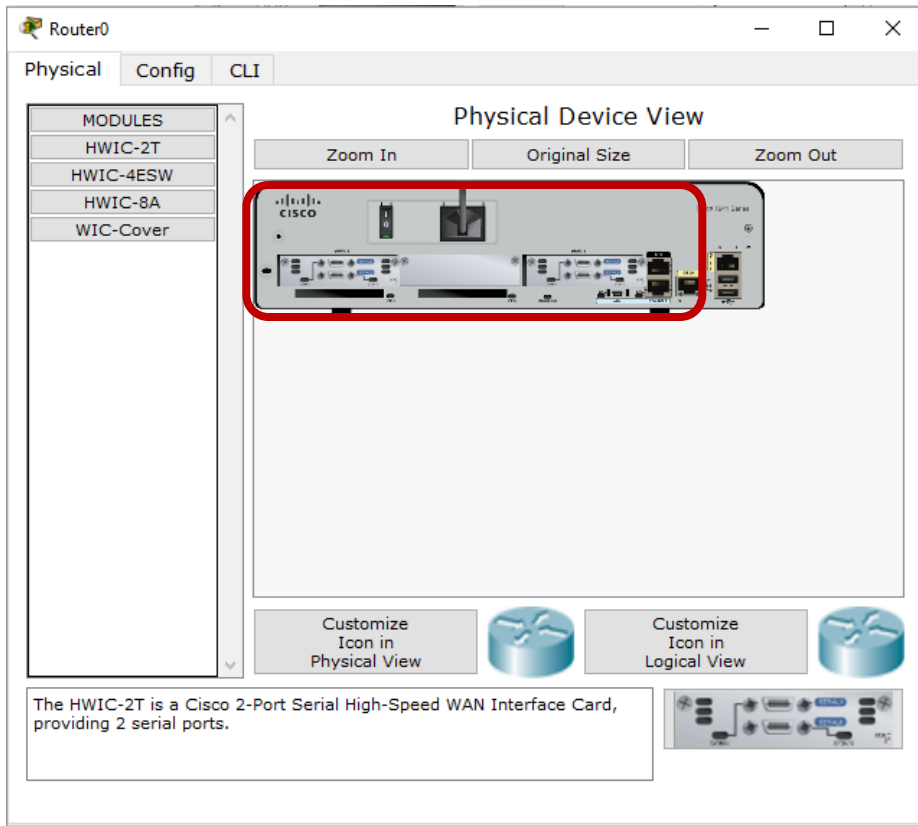
Step 11: Double click on the **Router0** icon and turn the router off by clicking the power off button



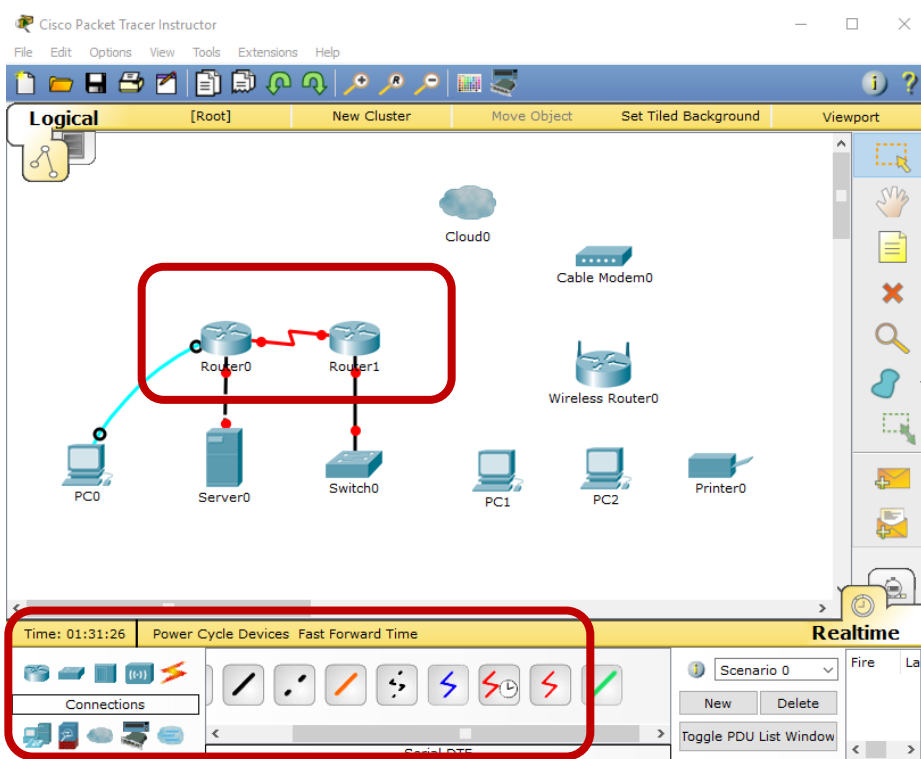
Step 12: Select **HWIC-2T** and add two **serial cards** to the slots (drag it to the empty slot)



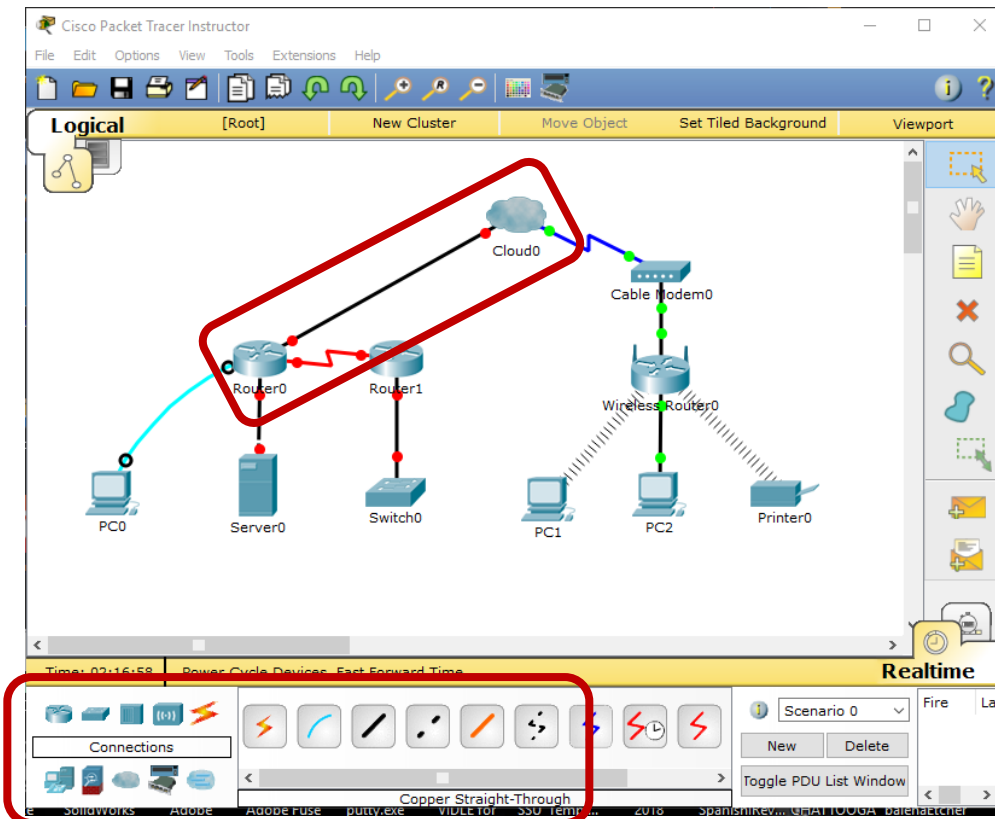
Step 13: Turn on **Router0** the power button then close the window and repeat the **steps 11 and 12** to add serial cards to **Router1**



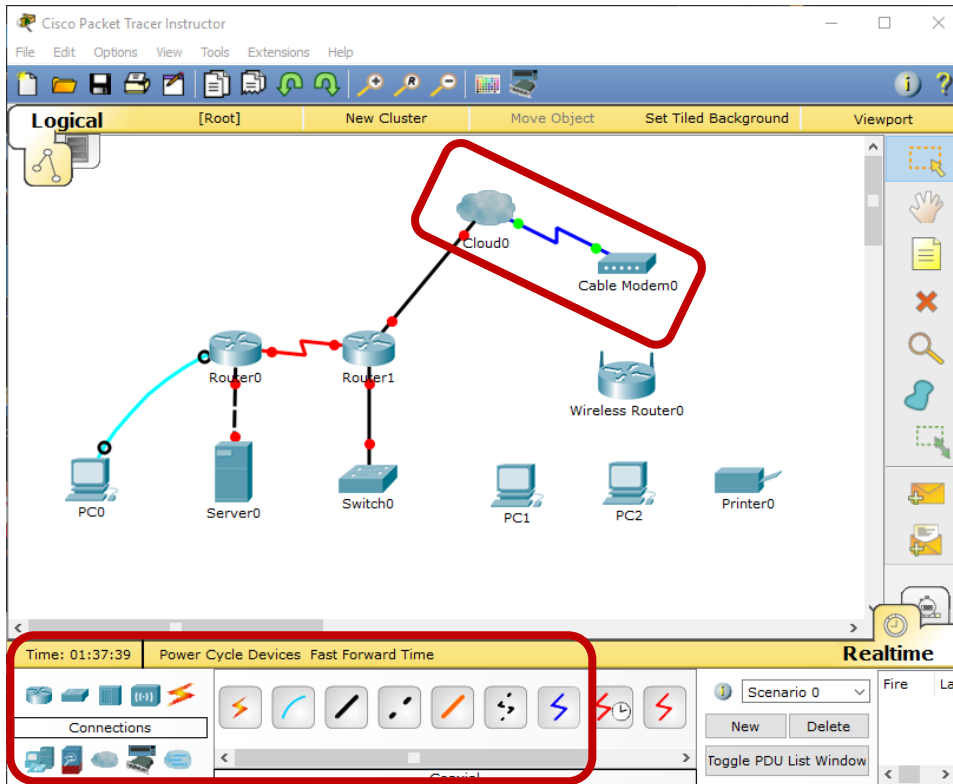
Step 14: Select **Serial DTE** cable to connect **Router0 (Serial0/0/0)** to **Router1 (Serial0/0/0)**



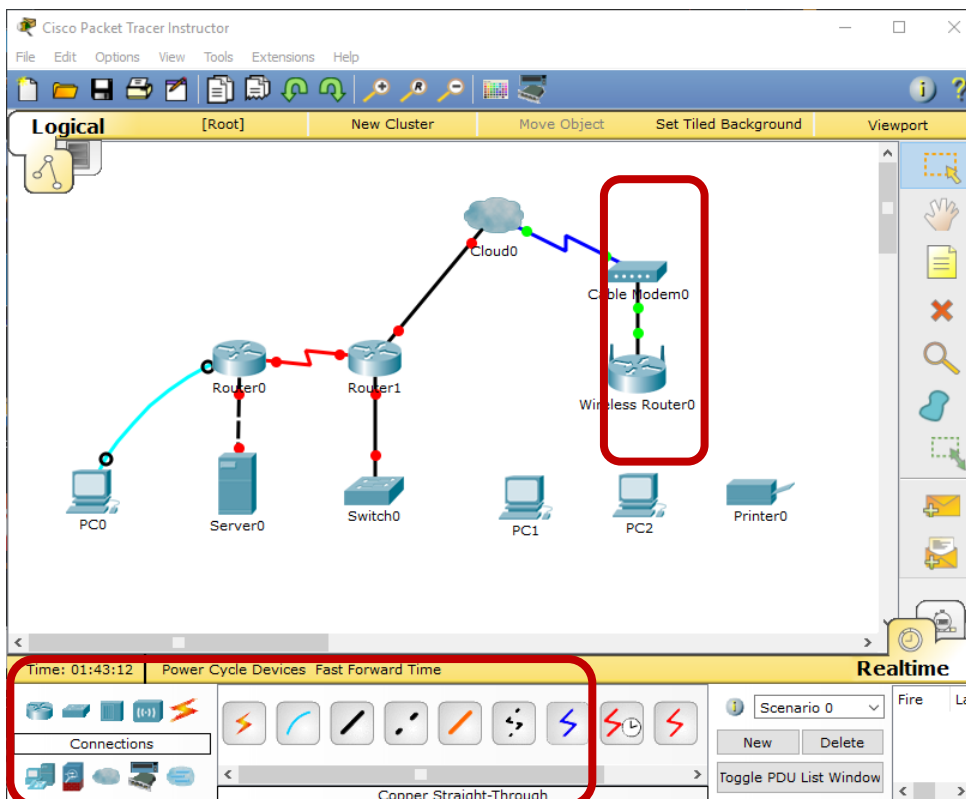
Step 15: Select **Copper Straight Through** to connect **Cloud0 (Ethernet6)** to **Router0 (GigabitEthernet0/1)**



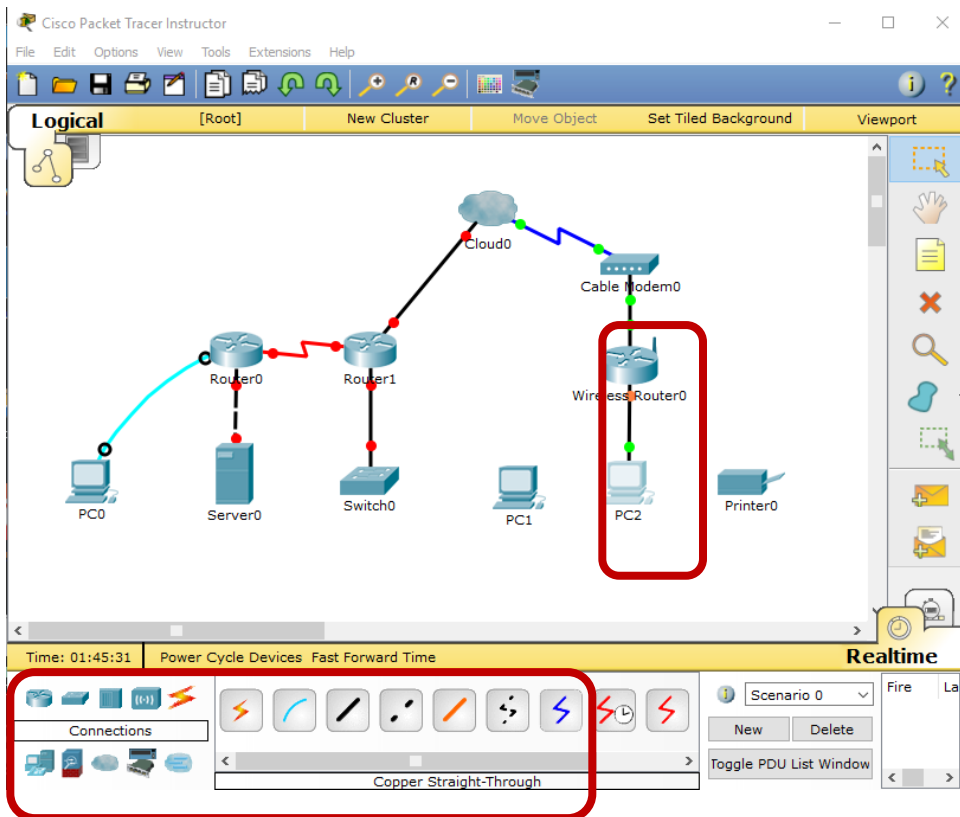
Step 16: Select **Coaxial** to connect **Cloud0 (Coaxial7)** to **Cable Modem0 (Port0)**



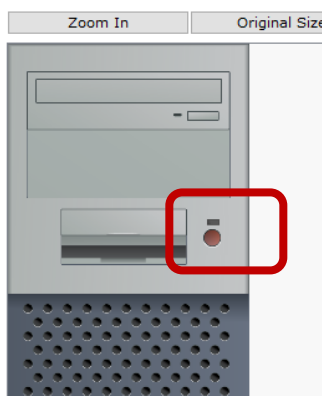
Step 17: Select Copper Straight Through to connect Cable Modem0 (Port1) to Wireless Router0 (Internet)



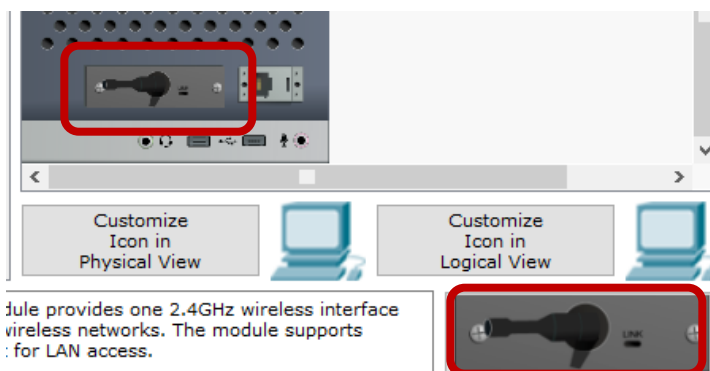
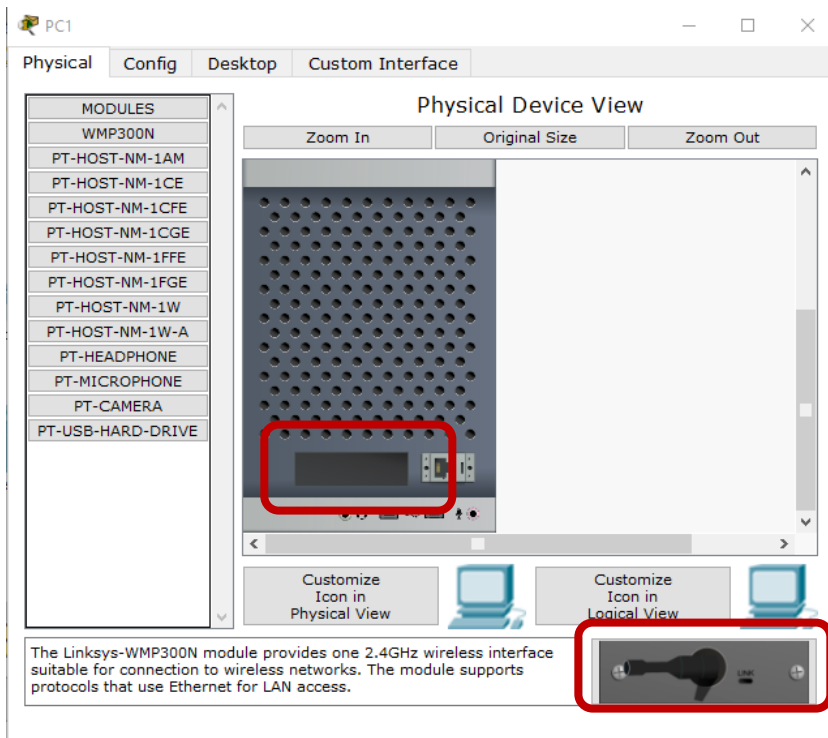
Step 18: Select Copper Straight Through to connect **Wireless Router0** (**Ethernet1**) to **PC2 (FastEthernet0/1)**



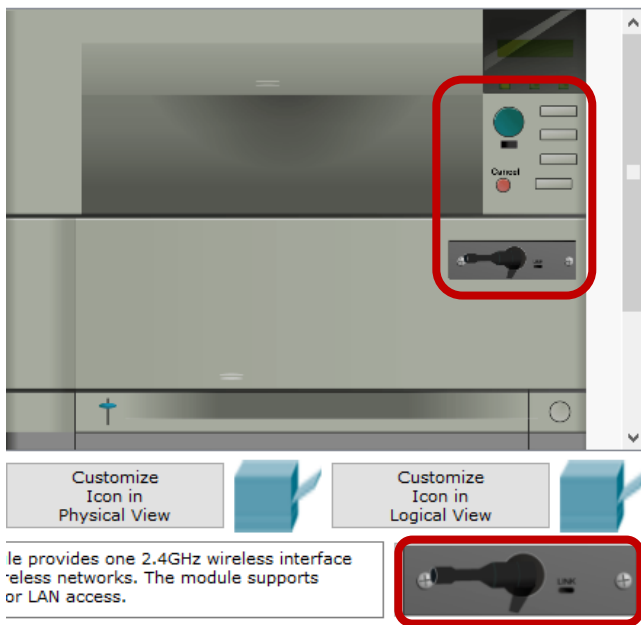
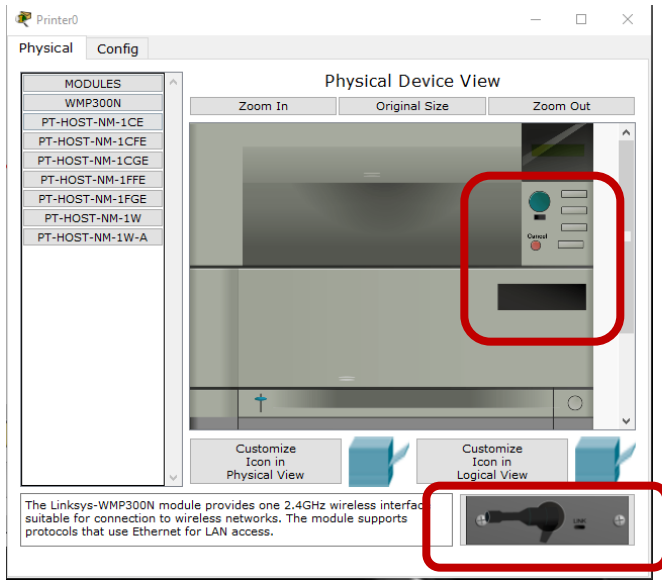
Step 19: Double click on **PC1** and turn off by pressing the power button then scroll down



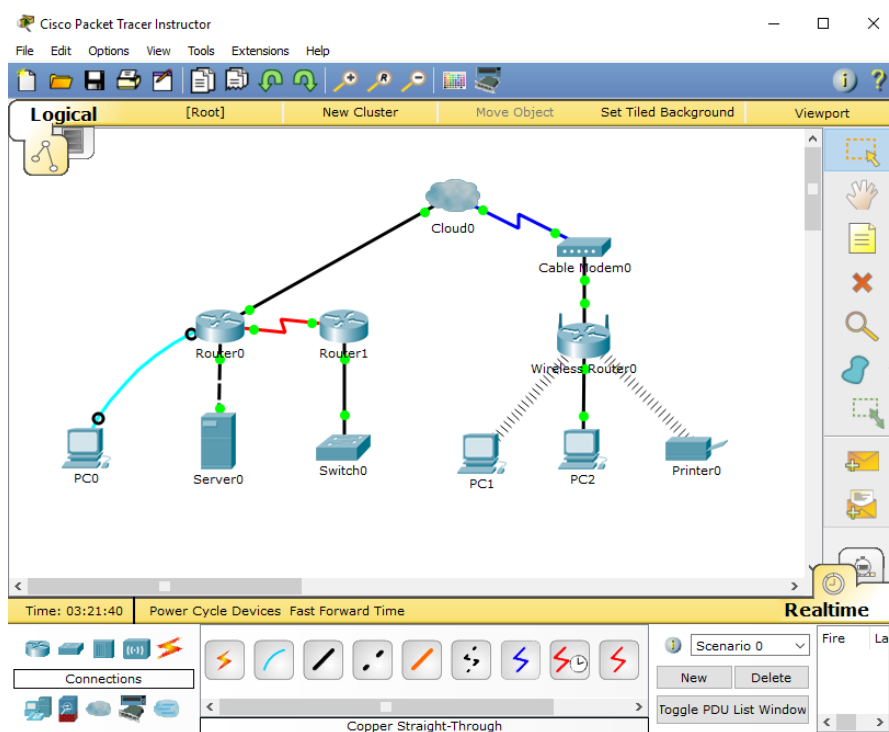
Step 20: Remove the network card by dragging the card off to the lower right of the window. Then select **WMP300N (wireless adapter)** and drag the adapter to the slot on the PC1 (this is the slot of the network card you just removed). Turn on the PC on.



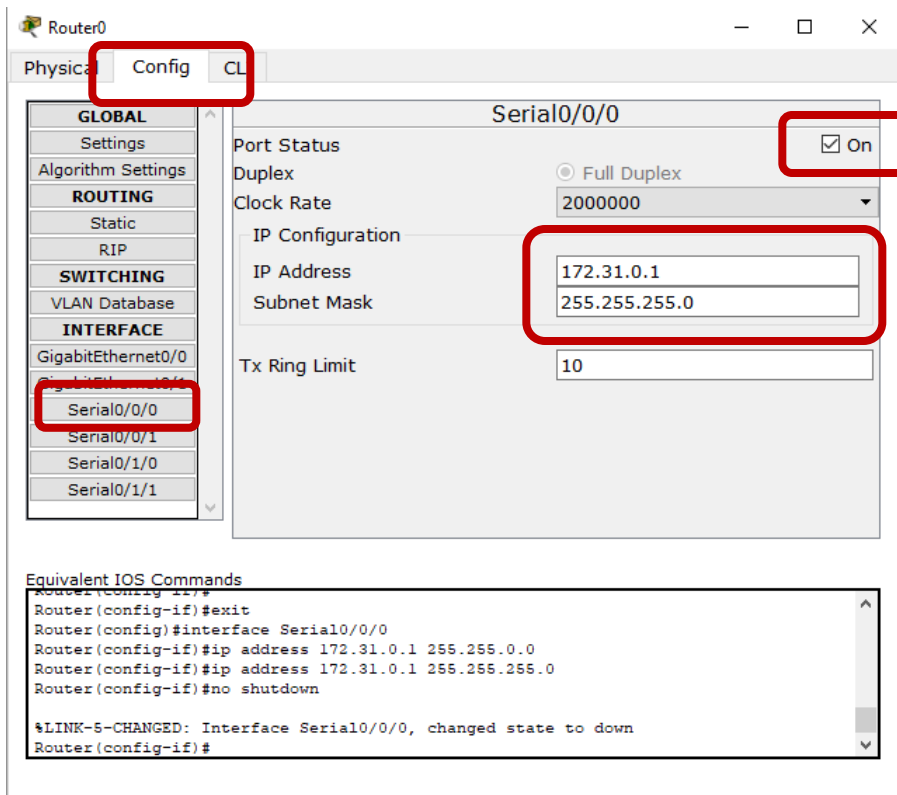
Step 21: Double click on **Printer0** and turn off by pressing the power button then remove the network adapter by dragging it to the lower right of the workspace. Then select **WMP300N (wireless adapter)** and drag the adapter to the slot on the Printer (this is the slot of the network card you just removed). Turn on the Printer on.



Step 22: At this point, the workspace should be similar to the figure below



Step 23: To configure Router0 double click on **Router0** the select **Config** tab and select **Serial0/0/0**. Enter the IP address and subnet mask (/24 means 255.255.255.0) provided in the table below and make sure the **On** checkbox is checked



Step 24: Repeat step 22 to configure remaining devices according to the table provide. PC1 and Printer do not require any configuration since automatically will be assigned an IP address. PC2 must be changed from **Static** to **DHCP** to automatically obtain an IP address.

Step 23: Configure the devices according to the table below:

Device	Interface	IP address	Connects to
Cloud0	Ethernet6	NA	Router0 (GigabitEthernet0/1)
	Coaxial7	NA	Cable Modem0 (Port0)
Cable Modem	Port0	NA	Cloud0 (Coaxial7)
	Port1	NA	Wireless Router0 (Internet)
Router0	Console	NA	PC0 (RS232)
	GigabitEthernet0/0	10.0.0.1/24	Server0 (GigabitEthernet0/0)
	GigabitEthernet0/1	192.168.2.1/24	Cloud (Ethernet6)
	Serial0/0/0	172.31.0.1/24	Router1 (Serial0/0/0)
Router1	Serial0/0/0	172.31.0.2/24	Router0 (Serial0/0/0)
	GigabitEthernet0/1	172.16.0.1/24	Switch0 (FastEthernet0/1)
Switch0	FastEthernet0/1	NA	Router1 (GigabitEtherner0/1)

Server0	FastEthernet0	10.0.0.254/24	Router0 (GigabitEthernet0/0)
Wireless Router0	Internet	192.168.2.2/24	Cable Modem0 (Port1)
	Ethernet1	192.168.1.1/24	PC2 (FastEthernet0)
PC0	RS232	NA	Router0 (Console)
PC1	Wireless Adapter		Wireless Router
PC2	FastEthernet0	192.168.1.102	Wireless Router0 (Ethernet1)
Printer	Wireless Adapter		Wireless Router

Step 25: Physical Layout of the SOHO network:

